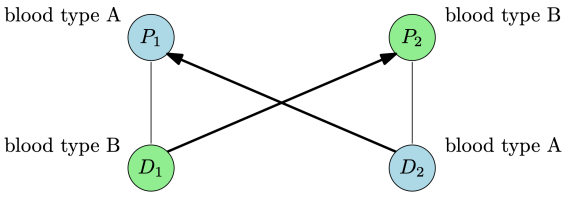
DS 574
Fall 2026

Kidney Exchange Worksheet
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Kidney Exchange

A patient–donor pair consists of a patient who needs a kidney and a willing donor who cannot donate to that patient because they are incompatible. In a two-way kidney exchange, two such pairs exchange donors: each donor gives to the other pair's patient, provided both donations are compatible.



Here P_i is patient i and D_i is their paired donor. The thin vertical lines identify the original pairs; the arrows show the cross donations $D_1 \to P_2$ and $D_2 \to P_1$.

Kidney exchange problem. Given patient–donor pairs and which donations are compatible, choose feasible exchanges. Each patient receives at most one kidney, and each donor gives at most one. We study exchanges without money, first using a housing-market analogy and then restricting to two-way exchanges to maximize the number of patients matched.

How would one design a centralized mechanism for kidney exchange, where incompatible patient–donor pairs can register and be matched with others?

(a) **Who are the agents?** Patient–donor pairs, or hospitals, or both?

(b) **What's private?** What can they misreport, and how?

(c) **What's the objective?** Whose welfare, measured how?

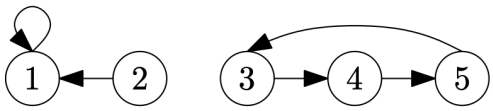
(d) **What's feasible?** Which exchanges can be carried out?

Idea #1: Top Trading Cycles

Consider the housing allocation problem defined by Shapley and Scarf: there are n agents, and each initially owns one house. Each agent has a strict total ordering over the n houses, and need not prefer their own over the others. How can we reallocate the houses to make the agents better off?

Top Trading Cycles (TTC). While agents remain:

- Each remaining agent points to the owner of their favorite remaining house. This induces a directed graph on the remaining agents in which every vertex has out-degree 1.
- The graph has at least one directed cycle. Self-loops count.
- Execute every cycle: each agent gets the house owned by the agent they point to.
- Delete these agents and their houses, and repeat.



Find the cycles in this iteration. Who receives which house? What happens to agent 2?

Why does the algorithm terminate? Why does each agent weakly prefer their allocation to their own house?

With strict preferences, TTC is DSIC (stated without proof).

Modifications for Kidney Exchange

The housing model is a useful starting point, but its assumptions matter. A TTC cycle can involve arbitrarily many agents. Here we restrict feasibility to pairwise exchanges, and model a patient as indifferent among compatible donors: utility is 1 if matched and 0 otherwise. These are modeling assumptions, not a claim that every kidney or every exchange program is identical. Patients without paired donors and donor-initiated chains require an extended model.

The lesson is to match the mechanism to the feasible exchanges and the information agents can hide. Coalition stability in a housing market does not establish truthfulness in a different reporting model.

Idea #2: Use a Matching Algorithm

The vertex set V consists of incompatible patient–donor pairs. An undirected edge $\{i, j\}$ means that donor i is compatible with patient j *and* donor j is compatible with patient i. A matching is a set of edges with no shared endpoints. Each edge implements two transplants, so maximizing the number of edges also maximizes the number of matched patients in this pairwise model.

Each agent i has a true set E_i of incident exchange edges and may report any subset $F_i \subseteq E_i$, but cannot invent a feasible exchange. The vertices and their priority order are fixed. The reported graph includes an edge only when both endpoints report it. Utility depends only on whether i is matched, not on which compatible donor they receive.

If two maximum matchings cover the same vertices, can any patient strictly prefer one to the other in this model? What if they cover different vertices?

Priority Tie-Breaking

Fix the vertex order $1, 2, \ldots, n$ before reports arrive. First maximize matching cardinality; among those matchings, prefer matching vertex 1, then vertex 2 subject to the first decision, and so on.

1. Let $\mathcal{M}_0$ be the set of maximum-cardinality matchings of the reported graph.
2. For $i = 1, \ldots, n$, let Z_i be the matchings in $\mathcal{M}_{i-1}$ that match i. If Z_i is nonempty, set $\mathcal{M}_i = Z_i$; otherwise set $\mathcal{M}_i = \mathcal{M}_{i-1}$.
3. Return any matching in $\mathcal{M}_n$.

Each family remains nonempty. All final matchings cover the same vertices, so the remaining choice of edges is irrelevant to utilities.

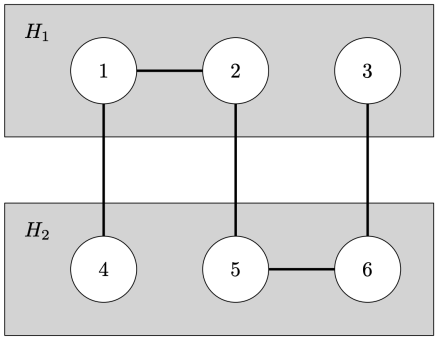
Example. On the path $1 - -2 - -3 - -4 - -5$, list the maximum matchings and apply priority $1, 2, 3, 4, 5$.

The priority mechanism is DSIC in this edge-hiding model (proof in the extension).

Hospital Incentives

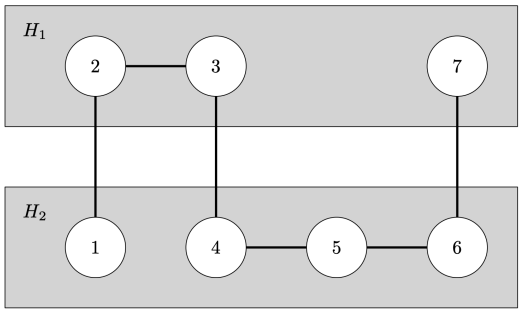
Hospitals may control which patient–donor pairs enter the exchange. A hospital's utility is the number of its own patients matched, including internal exchanges among pairs it withholds. Society counts all matched patients. This changes both the agent and the deviation: a hospital can hide vertices, not merely edges incident to one vertex.

The Need for Full Reporting



Compare internal matching first with reporting every pair.

Hiding Patients



H_1 owns $\{2, 3, 7\}$; H_2 owns $\{1, 4, 5, 6\}$. Find a maximum matching under full reporting. Now let H_1 withhold 2 and 3 for an internal exchange. What can the central exchange match? Count H_1's matched patients, including the internal exchange.

Which part of the individual-pair reporting model changed?

Extensions

The following questions develop the guarantees and alternative feasibility models.

Why Hiding Edges Cannot Help

Fix other reports and suppose vertex i is unmatched under a full report. Let M be the chosen matching. If i hides incident edges, does M remain feasible? Can a newly selected matching beat M under the fixed cardinality-then-priority comparison?

Coalitions and the Core

Core: which agents, which deviation? Return to the housing model, where every agent initially owns one house. An allocation assigns each house to exactly one agent. A coalition is a nonempty group of agents; it may redistribute only its members' original houses, not the houses assigned to them by the proposed allocation.

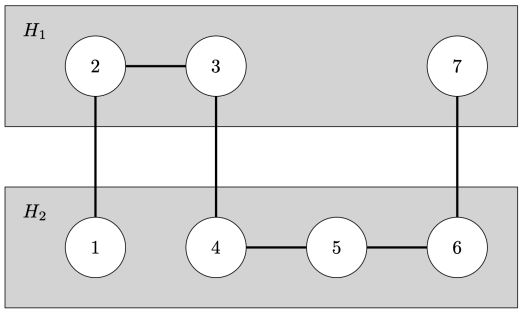
Strong core. A coalition weakly blocks an allocation if its members can redistribute their original houses so that every member weakly prefers their new house to their house in the allocation, and at least one member strictly prefers theirs. The strong core consists of allocations with no such coalition. We use this convention, rather than requiring every member of a blocking coalition to improve strictly.

Let N_j be the agents leaving TTC in round j.

For a proposed blocking coalition, consider its earliest departing members. If none may be worse off, which houses must they receive? Follow their ownership cycle, then remove it and continue.

For uniqueness, consider the first TTC round in which another allocation differs. Can one cycle block that allocation?

Hospital Impossibility



Assume the mechanism always maximizes matching cardinality in the reported graph. A hospital can hide pairs and match those pairs internally. H_1 owns $\{2, 3, 7\}$; H_2 owns $\{1, 4, 5, 6\}$. One patient is unmatched under full reporting. Handle both cases: that patient belongs to H_1 or to H_2. For each, give a withholding report and count the hospital's matched patients before and after. Does the argument depend on tie-breaking?

Two-Way versus Three-Way Exchange

A directed arc $i \to j$ means donor i can donate to patient j. Draw the compatibility graph on three pairs with precisely the arcs $1 \to 2$, $2 \to 3$, $3 \to 1$. Compare the maximum number of matched patients using only two-way exchanges and using a three-way exchange.